

Mitesh Kumar Anshu

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SUMMARY

Full-stack engineer specialising in backend reliability, transactional data integrity, and role-based system design. Builds end-to-end features across React, Node.js, and PostgreSQL — from RBAC middleware and PL/pgSQL stored procedures to deployed, live applications. Strong foundation in data structures, OOP, and concurrency-safe schema design.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL

Frontend: React.js, Vite, Zustand, Material UI, Recharts, HTML, CSS

Backend: Node.js, Express.js, REST APIs, JWT Auth, RBAC Middleware, Helmet, CORS

Databases: PostgreSQL (Transactions, Row-Level Locking, Stored Procedures, Trigram Indexes), MongoDB

Tooling: Git, GitHub, Postman, VS Code

Deployment: Vercel, Render

PROJECTS

CircuLib — Library Circulation & Operations Platform React · Node.js · PostgreSQL

[GitHub](#) · [Live](#)

- Built an end-to-end, role-based library platform with three access-controlled portals (Admin, Librarian, Student) using React (Vite), Express, and PostgreSQL.
- Designed a layered backend with routes, JWT authentication, RBAC middleware, and controllers.
- Centralised database access with connection pooling, retries, and transaction helpers.
- Enforced circulation integrity via PL/pgSQL stored procedures (*sp_checkout_book*, *sp_return_book*) — loans, inventory, fees, and alerts update atomically in one transaction.
- Eliminated double-loan race conditions with a unique partial index (*ux_loans_copy_active*) and *SELECT FOR UPDATE* row-level locking at the DB layer.
- Developed 12 API domains: auth, RBAC, catalog, copies, circulation, reservations, reviews, wishlist, announcements, reports, fees, and global search.
- Implemented fuzzy search via PostgreSQL trigram indexes with parallel queries across catalog, members, loans, and announcements.
- Built analytics dashboards with Recharts backed by PostgreSQL views for aggregated reporting.
- Deployed React frontend on Vercel and Node.js backend on Render; configured CORS policies, environment variables, and JWT auth across both services.

Smart Billing & Discount Engine Java · OOP · Design Patterns

[GitHub](#)

- Applied the Strategy Pattern for interchangeable discount algorithms (percentage, fixed, BOGO) — billing logic remains closed for modification, open for extension.
- Designed *ShoppingCart* and *Item* with composition-based discount injection, keeping pricing logic modular and independently testable.
- Separated *cart* and *discount* modules to cleanly decouple domain models from pricing behaviour.

EDUCATION

Roorkee Institute of Technology, Uttarakhand — B.Tech, Computer Science Engineering | 2021 – 2025

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Object-Oriented Programming, Computer Networks

ACHIEVEMENTS

- Solved 100+ problems on LeetCode and GeeksforGeeks covering arrays, trees, graphs, and OOP design — consistent practice with a focus on problem-solving patterns.
- Produced full technical documentation for CircuLib: ER diagrams, stored procedure flows, auth flow, and API reference — enabling faster onboarding and technical review.